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SheepDoctor: A knowledge graph enhanced large language model for sheep disease diagnosis

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ABSTRACT

Traditional methods for diagnosing sheep diseases often fail to provide timely and accurate results, particularly in emergency situations, and public awareness of these diseases remains limited. This hinders effective prevention and control efforts. To address these challenges, a domain-specific large language model, SheepDoctor, was developed for sheep disease diagnosis. A comprehensive question-and-answer (Q&A) dataset was constructed using prompt techniques, resulting in 5987 samples covering 207 sheep diseases. This dataset included detailed symptom descriptions, treatments, and related information, which were also structured into a knowledge graph. The pre-trained LLaMA2–13B-Chinese model was fine-tuned using Low-Rank Adaptation (LoRA), with integration of the knowledge graph to enhance its diagnostic capabilities and response accuracy. Evaluation using BLEU, ROUGE, and BERTScore metrics demonstrates that SheepDoctor outperforms general-purpose models such as GPT-4o and Kimi on sheep-related diagnostic tasks. The proposed method exhibits strong domain expertise and holds significant potential for improving the efficiency and reliability of sheep disease diagnosis.

1. Introduction

Diagnosing and preventing sheep diseases is a significant challenge for farmers and veterinarians. Traditional diagnostic methods involve on-site veterinary visits and examinations, which has several problems, including limited access to veterinarians, delayed medical consultations, and low diagnostic efficiency due to farmers' limited expertise. Although the variety of sheep diseases is small compared to human diseases, the high similarity in symptoms complicate the diagnostic process. Since the release of ChatGPT [1], Large Language Models (LLMs) have gained considerable attention worldwide. These models possess advanced natural language understanding and generation capabilities, allowing them to learn from extensive textual data and grasp complex language patterns and semantics. This provides a new solution for sheep disease diagnosis.

LLMs exhibit powerful Emergent Abilities [2] by fine-tuning or giving the model more domain-relevant datasets. This is especially evident in the field of medical diagnosis, where LLMs have made significant progress. By integrating LLMs with healthcare data, these models offer valuable tools for accelerating discoveries and enhancing health outcomes. Notably, Large Language Models, such as ChatGPT, have demonstrated excellent performance in answering questions related to ophthalmology [3], surgery [4], and genetics [5] that surpasses that of the human, approaching the proficiency of experts.

Current medical diagnostic systems based on LLMs primarily serve three functions. First, they enable doctor-patient dialogue. For instance, Xiong et al. [6] trained ChatGLM using traditional Chinese medicine and human disease Q&A data to develop a Chinese doctor-patient dialogue model. Li et al. [7] fine-tuned the LLaMA [8] with real doctor-patient dialogue data scraped from online medical websites to create an English-language dialogue model. Zhang et al. [9] trained HuaTuoGPT, while Wang et al. [10] created BenTsao, both of which fine-tuning the LLaMA to produce Chinese doctor-patient dialogue models. Second, these systems can generate diagnostic reports. For instance, Luo et al.

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[11] trained Taiyi using bilingual biomedical data (Chinese and English) to perform intelligent Q&A and report generation in the biomedical domain. Finally, these systems can integrate multimodal data for disease imaging diagnostics. For instance, Thawakar et al. [12] developed a multimodal framework that integrates chest X-ray images with textual data to enhance medical diagnosis through vision-language models. In addition to multimodal applications in the medical field, the excellent performance of LLMs in other image recognition domains [13], such as vehicle brand recognition, explains the potentials and limitations of multimodal LLMs in dealing with tasks in the domains with specialized visual features, and also provides inspiration and references for extending them to agricultural recognition tasks.

Most existing research predominantly focuses on human disease diagnosis, while diagnostic methods in agriculture, including animal diseases, largely remain limited to traditional expert consultations. A few studies have explored the use of LLMs and knowledge graph-based methods in agriculture. ShenNong [14], developed by China Agricultural University, addresses four key areas: breeding, farming, planting, and remote sensing meteorology. This model incorporates specialized agricultural models designed to tackle challenges such as seed selection, disease management, pest identification, and natural disaster warnings. And FanLi [15], as China's first agricultural large model dedicated to the fisheries sector, processes multimodal data including text, images, videos, and audio to establish a comprehensive knowledge base for fisheries. It provides more accurate and holistic AI-driven support for aquaculture, facilitating health management and disease early warning, while promoting intelligent and efficient fisheries production. Hoang et al. [16] introduced the LiteralKG model, which enhances the efficiency of animal disease diagnosis by integrating literal information and graph structures from knowledge graphs into a unified representation. Moreover, Yang et al. [17] constructed a sheep disease knowledge graph by scraping relevant data and applied the BERT [18] to enable search and response functionalities for sheep disease diagnosis. Currently, there is a significant lack of large models and related research specifically focused on the diagnosis of sheep diseases. This gap limits the ability to provide accurate, efficient, and scalable solutions for disease management in sheep farming.

To solve these problems, the large model was augmented with a knowledge graph to further improve its ability to understand the expertise in the field of sheep diseases. First, aa comprehensive data was collecting professional data on 207 sheep diseases, and then organized into a Q&A dataset and knowledge graph (KG). Then, the pre-trained LLaMA2–13B-Chinese was fine-tuned using this dataset to enhance its ability to answer queries related to sheep diseases. Finally, the integration of the KG further strengthened the LLM's ability to accurately interpret and reason over sheep disease information.

The remainder of this paper is organized as follows. Section 2 describes the dataset used in this work and its construction process, followed by a detailed explanation of our proposed method. Section 3 presents the experimental results. A discussion of the results is provided in Section 4. Finally, conclusions are drawn in Section 5.

2. Materials and methods

This section provides a detailed description of the materials and methods employed in this study. First, the overall research framework is outlined. Subsequently, the dataset construction process is described in detail. Finally, the fine-tuning approach for the large model and the knowledge graph enhancement method are elaborated separately.

2.1. The overall flow of constructing SheepDoctor

The overall construction flow of SheepDoctor is illustrated in Fig. 1. First, data is sourced from various authoritative references, including specialized books, reputable websites, and actual diagnostic cases. The data is preprocessed, and resulting in two distinct formats: one for

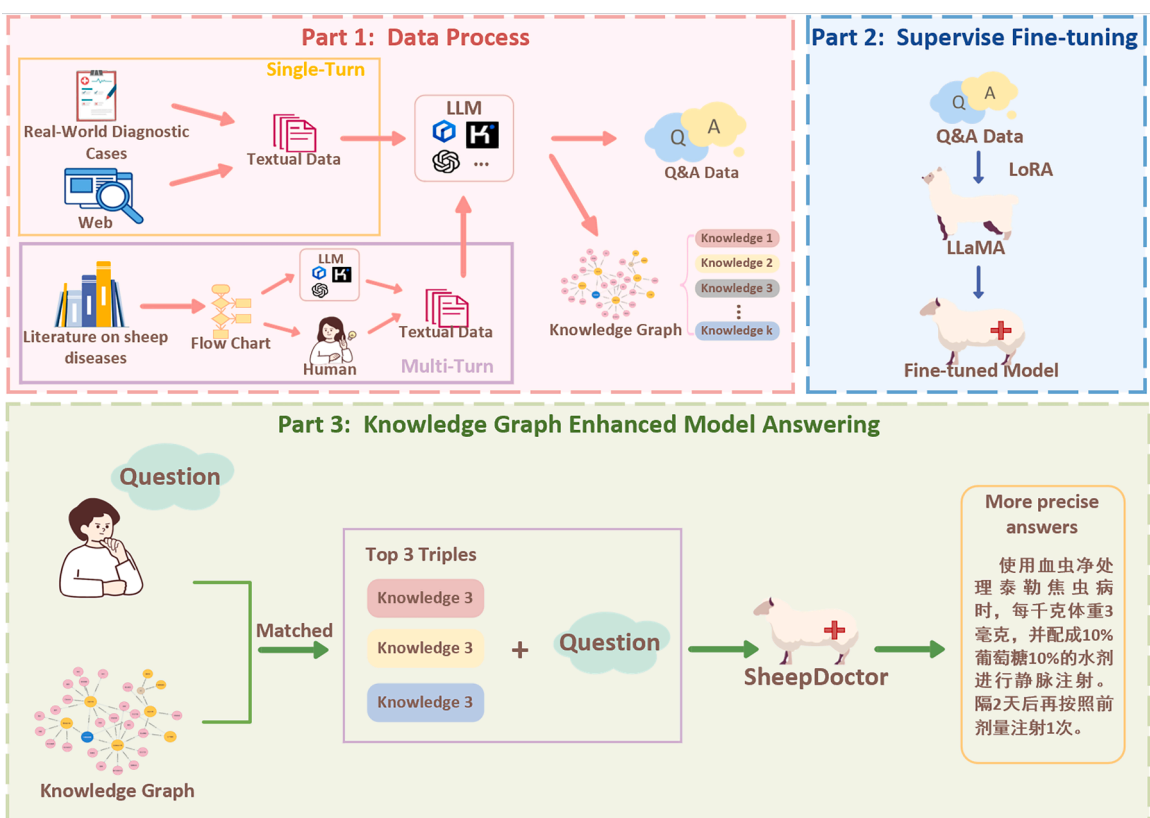


Fig. 1. The implementation framework of SheepDoctor.

constructing the knowledge graph in a ternary format, and the other for the Q&A format. Then, this specialized data is used to fine-tune a pre-trained LLM to creating a comprehensive model capable of addressing queries related to sheep diseases. Finally, the knowledge graph is used to enhance the model's responses, aligning them more closely with user inquiries, thereby ensuring the development of a reliable and precise diagnostic tool for sheep diseases.

2.2. Construction of the dataset

There are generally two methods for obtaining sheep disease data for training LLMs. One is extracted from existing general-purpose LLMs. While LLMs perform well on smooth dialogue, they often lack specialized knowledge in vertical domains, resulting in lower-quality sheep disease information compared to professional veterinary diagnostics. The other method is to gather data from actual veterinary diagnostics, which are highly specialized but may still be subjective. As a result, the responses generated by the model after training tend to be concise and lack detail; veterinary replies are professional but less interactive than those from LLMs. Therefore, a hybrid approach was adopted, combining real-world data with LLMs to develop a more comprehensive dataset for model training.

The real-world data is sourced from open books on sheep diseases [19], the animal husbandry website [20], and unpublished case records of sheep diseases diagnoses. The dataset includes 207 sheep diseases which categorized into 10 types, along with 1152 symptoms. Additionally, it provides information on 487 therapeutic drugs, including their dosages and usage instructions.

The real-world data is provided to the existing generic LLM such as LLaMA2–13B-Chinese, in different formats to generate sheep disease instruction data, similar to the Alpaca [21] format. Different prompting methods are designed for introducing sheep diseases, treatments, drug dosages, and other relevant information. These methods either assign roles to the commands or alter the phrasing of questions to diversify the dataset. The outputs are rewritten based on the original text, with the prompting methods detailed in Table 1.

Diagnosing sheep diseases based on symptoms is a complex process that often requires multiple rounds of question-and-answer interactions between a veterinarian and the farmer. Real-world diagnostic data derived from symptoms are typically represented as flowcharts, which present challenges for current LLMs, as they struggle to interpret this visual data effectively. As a result, human experts must first grasp the intricate diagnostic process before guiding the models on how to generate actionable instructions. To ensure that our diagnostic model mimics the approach of a real veterinarian, a sample of the prompts used for multi-round Q&A dialogues is provided in Table 2.

By transforming raw text and flowcharts into a format suitable for LLM training, as demonstrated in the prompted methods of Table 1 and Table 2, it is ensured that the data accurately reflects the complexities of the real diagnostic process. This careful preparation enables the model to generate dialogue content that is detailed, rich, and consistently structured.

Table 1
A sample of prompts generated based on real sheep disease information.

Number	Type	Prompt
1	Basic information on sheep diseases	请按照以下要求将给定的中文文本改写成问题与答案格式: 待改写文本:{{Original Text}} 改写要求: 保持原文的大体结构和主要信息不变, 但可以对语气,提问方式,病情描述等进行调整, 以符合实际兽医就诊场景. 改写后的文本应简洁明了, 易于理解, 同时保持原文的准确性和专业性. 请确保改写后的文本符号,格式和英文字母等符合实际医疗要求, 避免产生歧义或误导. 请确保改写后的文本既符合医疗行业的规范, 又能准确传达原文的意思.
2	Drug dosage information	请按照以下要求将给定的中文文本改写成问题与答案格式: 待改写文本:{{Original Text}} 改写要求: 问题方式可以对语气,提问方式,病情描述等进行调整, 以符合实际兽医就诊场景. 改写后的文本药品名称,剂量,使用方法,单位等应保持原文的准确性和专业性. 为保证问题及回答的多样性, 可添加一些关于羊品种, 年龄, 性别, 养殖环境之类的细节描述.

2.3. Fine-tuning

Training a large language model for domain-specific question answering generally involves four steps [22]: pretraining, fine-tuning, reward modeling, and reinforcement learning. The latter two are collectively referred to as reinforcement learning with human feedback (RLHF). The pretrained model is trained on extensive data, capturing both syntax and semantic information, which provides rich feature representations for downstream tasks. Since training a large language model from scratch demands significant computational resources and time, fine-tuning a pretrained model allows for effective performance using relatively less domain-specific data and resources, thereby facilitating the completion of specific domain tasks.

The pre-trained model used in this study is LLaMA2–13B-Chinese. Because the original LLaMA2 model has relatively weak alignment for Chinese, and our dataset is entirely in Chinese. To address this, a pre-fine-tuned Chinese version of the model was adopted and fine-tuned with domain-specific data for sheep disease diagnosis. This approach ensures that the model not only better understands and processes Chinese but also grasps the specialized terminology and diagnostic logic inherent in the domain-specific data.

In the supervised fine-tuning phase (SFT), we utilized the methodology recounted in Section 2.2 to construct a dataset including 5978 entries. This dataset is essential for enabling the model to simulate realistic Q&A interactions, resembling those between veterinarians and farmers, while maintaining the logical consistency and coherence established during pretraining. To achieve efficient parameter tuning with minimal computational overhead, the low-rank adaptation (LoRA) method [23] is employed. By selectively updating only a small subset of the model's parameters, particularly those associated with low-rank matrices, LoRA significantly reduces the computational and memory demands. This targeted approach to parameter updating not only mitigates the risk of overfitting on smaller datasets but also enhances the model's interpretability, providing clearer insights into how the model learns and adapts to new tasks.

Let W represent the original weight matrix in the pretrained model. LoRA is implemented by adding a learnable low-rank matrix decomposition Δ_W to the weight matrix W of some layers, as shown in Eq. (1):

$$W' = W + \Delta W \tag{1}$$

Here, Δ_W is derived from the product of two low-rank matrices, as represented in Eq. (2), where d denotes the dimension and r represents the rank of the low-rank matrix. This decomposition allows the large-scale parameters that LoRA must learn to be transformed into smaller low-rank matrices, thereby reducing computational overhead:

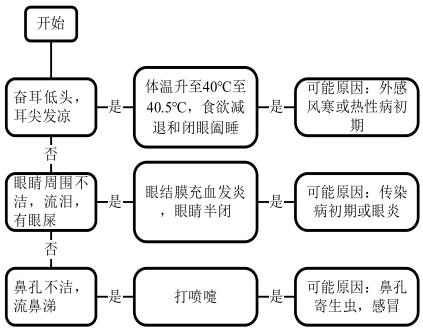
$$\Delta W = AB(A \in R_{d \times r}, B \in R_{r \times d}) \tag{2}$$

To enhance the impact of the low-rank matrix on the model, LoRA introduces a training factor α, resulting in the following transformation of the low-rank matrix formula, as shown in Eq. (3):

$$\Delta W = \frac{\alpha}{r} AB \tag{3}$$

Additionally, the dropout technique can be incorporated during training to prevent overfitting, as shown in Eq. (4):

Table 2
A sample of multi-turn Q&A data generation prompts.

Prompt-1	Converted text	Prompt-2
以下为根据症状诊断羊类疾病的一个流程图.  流程图中涉及"是否"判断, 请你从流程图的"开始"位置开始, 遍历每一条判断直到最后判断出可能的疾病.	1.**耳尖发凉,羊的体温升至40 °C至40.5 °C, 食欲减退和闭眼阖睡**:这可能是外感风寒或热性病初期的症状. 2.**眼睛周围不洁和流泪,眼结膜充血发炎,眼睛半闭**:这可能是由于眼部感染或刺激导致的. 3.**鼻孔不洁和流鼻涕,打喷嚏**:这可能表明羊的鼻孔有寄生虫感染或感冒.	请将以下内容转换为多轮问答格式: 原始文本:{{Converted text}} 要求: 1.模拟养殖户与兽医的交流语气, 养殖户提问, 兽医回答或反问. 2.人类对于症状-疾病提问一般不会描述全部症状, 因此, 模拟兽医回答时应适当生成一些引导性话语或反问. 3.主旨内容请保持与原文相同的准确性与专业性

$$\Delta W = \frac{\alpha}{r}\ \mathrm{Dropout}(AB) \tag{4}$$

2.4. Knowledge graph enhancement

Knowledge graphs are primarily represented as triples, capturing numerous entities and their interrelationships. This rich background knowledge equips the model to generate more accurate and detailed responses. In this paper, we extracted disease and symptom information in tebular form from websites and professional diagnositic cases. Through manual extraction, we further collected treatment plans associated with diseases, ultimately constructing a knowledge graph containing 1934 triples. An example of triples related to sheep diseases is presented in Table 3. Fig. 2 shows a partial visual representation of the constructed knowledge graph, where the blue node (disease type) and the green node (disease name) are shown as species relationships. The entity of disease name is connected to the pink node (disease symptoms) and the brown node (treatment method), and the name of the treatment method entity is represented by a number, because different diseases may correspond to the same drug (yellow nodes) with identical usage methods, and a number can be used to establish a relationship between the different treatment options corresponding to different diseases. Among them, drugs also contain attributes such as dosage, method of use, frequency of use, etc., to ensure the accuracy of the model's recommended treatment programs.

By calculating the similarity between a question and the knowledge graph triples, the most relevant triples for each question can be retrieved. This matching mechanism ensures that the model references the most pertinent knowledge when answering questions, improving accuracy and reliability. The methodology adopted in this paper for processing user queries involves aligning them with sheep disease-related triples. To achieve this, the Jieba [27] tokenizer was adopted.

Table 3
Example of sheep disease triples.

Head Entity	Relation	Tail Entity	Discription
羔羊肠痉挛	肌内注射	10 %安乃近3毫升	疾病与药物之间的关系
流产	表现为	努责	疾病与症状之间的关系
口蹄疫	属于	传染病	疾病与疾病类型之间的关系
普通山羊	繁殖年限	8年	羊只种类与养殖方法之间的关系
1月龄的羊	门齿数量	8颗	羊只年龄与特征之间的关系
保泰松	弃奶期	5天	药物与使用注意事项之间的关系
5 %碘醚柳胺油剂	屠宰前停药时间	15天	

This ensures that domain-specific terminology pertinent to sheep diseases is accurately identified and tokenized, which is crucial for capturing the precise intent of the user's query. Once the questions are segmented, the corresponding disease triples are matched, and this information, along with the original question, is formatted into prompts. These prompts are then fed into the model to elicit the final response. The matching mechanism are shown in Eq. (5):

$$\mathrm{cosine_similarity}(q_i,\ t_j) = \frac{q_i \cdot t_j}{\|q_i\|\ \ \|t_j\|} \tag{5}$$

The BERT is used to represent each word in the input question or triple as a vector. The entire sentence or triple is then represented by averaging these vectors or by using the classification token vector from the last layer of BERT. The vector representation of the question is denoted as q, q_i is the i th question, and the vector representation of the triple is denoted as t, t_j is the j-th triple. To measure the similarity between the question vector and the triple vector, the L_2 norm is used to compute the lengths of the vectors. The cosine similarity between the vectors is then calculated, which measures the similarity between them. The cosine similarity value ranges between [−1, 1], values closer to 1 indicate higher similarity between the vectors, meaning the question and the triple are more closely matched. An example of triple matching result and prompt templates is shown in Table 4. The final triples will be generated together with the questions by constructing a prompt template input model for the answers.

3. Results

This section provides a detailed discussion of the experimental results. The experimental parameters and evaluation metrics are firstly introduced, including BELU [24], ROUGE [25], BERTScore [26], and Number-Units-Matched. These metrics comprehensively assess the model's generation from three dimensions: language quality, semantic similarity, and domain specificity. Then, an ablation study is conducted for both quantitative and qualitative analysis of the proposed model. Finally, the proposed model is compared with common models such as LLaMA2–13B, ChatGPT-4o, and Kimi to further validate its superiority in the task of sheep disease diagnosis.

3.1. Experimental parameters and evaluation metrics

The experiments were conducted on a Linux Ubuntu distribution, featuring an Intel(R) Xeon(R) Gold 6130 CPU running at 2.10 GHz and equipped with 25 GB of RAM. The GPU utilized for the experiments is a V100 with 32 GB of memory. The Python version used is 3.8, while the

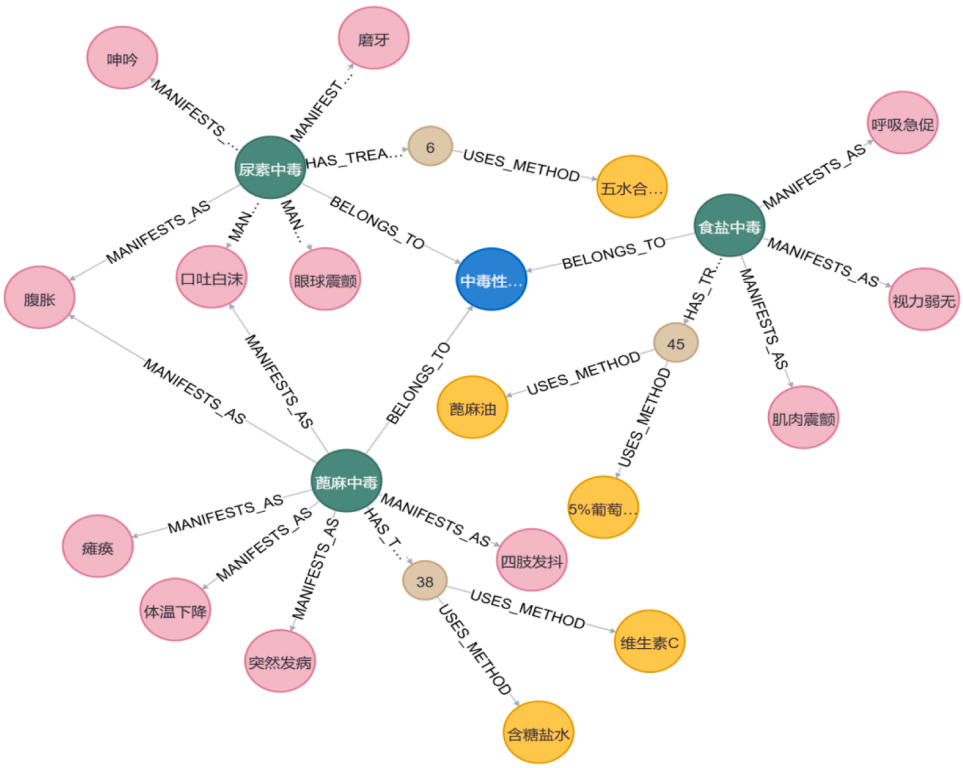


Fig. 2. Visualization of partial knowledge graphs.

Table 4
Sample of Triple Matching Results and Prompt Templates.

Number	Question	Triple	Prompt
1	在治疗重瓣胃阻塞时, 静脉注射5 %葡萄糖的量应该是多少∞	["重瓣胃阻塞", "静脉注射", "5 %葡萄糖400毫升"]	请仔细阅读以下非结构化信息, 若参考信息与问题相关, 请结合参考信息准确,专业地回答.
2	我使用了1 %伊维菌素来治疗羊的外寄生虫, 弃奶期需要多长时间∞	["1 %伊维菌素", "弃奶期", "15天"]	参考信息: {triples} 若信息与问题无关, 按你所知信息进行回答.
3	出血性肠炎的死亡率是多少∞	["出血性肠炎", "死亡率", "30 %"]	用户问题: {question} 请基于上述信息, 给出准确的回答:

CUDA version is 11.8, and the corresponding PyTorch version is 2.0.0.

The model is fine-tuned using the LoRA method, with the model training requiring approximately 26GB of GPU memory. The training parameters are as follows: learning rate: 0.00005, lora_alpha: 16, lora_rank: 8, lora_dropout: 0, epoch: 10, maximum sequence length: 1024.

3.2. Ablation experiments for quantitative analysis

To evaluate the effectiveness of the proposed model, we conduct ablation experiments using 395 data in Q&A format closely related to the triples for quantitative analysis. As shown in Table 5, Fine-tuned Model significantly outperforms LLaMA2–13B-Chinese in terms of BLEU-1 to BLEU-4, indicating that domain-specific fine-tuning substantially enhances the fluency and accuracy of text generation. However, the BLEU scores of SheepDoctor integration slightly decrease, likely due to the increased diversity in generated content, which may reduce the literal matching with reference texts. For example, while the reference text might state, "Intramuscular injection of Suxamethonium is typically used during surgery, at a dosage of 2–3 mg per kg body weight," the model's output could be phrased as, "Suxamethonium is the anesthetic used in cesarean sections, at a dosage of 2 mg/kg, administered intramuscularly." Despite this, SheepDoctor achieves superior performance on the ROUGE metrics (ROUGE-1: 58.11 %, ROUGE-2: 29.36 %, ROUGE-3: 56.65 %), demonstrating that KG significantly enhances semantic consistency and key information matching. Additionally, SheepDoctor outperforms the version without KG in both BERTScore (78.21 % vs. 78.15 %) and the Number-Units-Matching matric (70.03 % vs. 52.86 %), validating the positive impact of KG in improving entity understanding and domain knowledge coverage.

We further evaluated the generalization ability of the models on unseen problems. Unseen problems primarily involve diseases not included in the model's fine-tuning data, meaning the model has no prior knowledge of their specific symptoms or treatments, simulating scenarios where the model encounters new diseases. In particular, the Q&A data contained 33 diseases, totaling 77 data items. As shown in Table 6, the generalization ability of the LLaMA2–13B-Chinese is more

Table 5
The effect of with/without knowledge graph triples on model generation.

Type	BLEU-1(%)	BLEU-2(%)	BLEU-3(%)	BLEU-4(%)	ROUGE-1(%)	ROUGE −2(%)	ROUGE -L(%)	BERTScore (%)	Number-Units-Matched (%)
LLaMA2–13B -Chinese	22.25	14.70	10.62	7.76	8.79	1.11	8.71	67.96	34.61
Fine-tuned Model	41.35	34.57	30.18	26.50	33.67	14.12	32.93	78.15	52.86
SheepDoctor	40.29	33.35	28.55	24.30	58.11	29.36	56.65	78.21	70.03

limited, which suggests that the model struggles to effectively respond to new problems in the absence of domain fine-tuning and knowledge graph enhancement. In contrast, Fine-tuned model achieves slight improvement in precision, recall, and F1 value, indicating that domain fine-tuning enhances the model's ability to adapt to unseen problems to some extent, but the improvement is limited. In addition, the performance of SheepDoctor is significantly improved, with its precision, recall and F1 value reaching 80.37 %, 76.52 % and 78.15 %, respectively, and the matching rate of numerical units is also improved to 76.19 %, which suggests that the introduction of knowledge graph substantially enhances the model's ability to generalize to unseen problems, enabling it to more accurately capture key information and generate high-quality responses. In summary, the combination of domain fine-tuning and knowledge graph significantly improves the model's performance on unseen problems.

3.3. Ablation experiments for qualitative analysis

Table 7 shows the qualitative analysis of different models for a specific problem. While the LLaMA2–13B-Chinese mentions the general uses of codonopsis, aconite, and cinnamon, it fails to accurately identify their specific application in treating hypoglycemia and does not provide dosage information. This indicates that, without domain-specific knowledge, the base model struggles to generate accurate and detailed responses. In contrast, Fine-tuned model demonstrates an improved ability to recognize the medicinal value of these herbs and provides dosage information. However, its response still deviates from the reference answer, as it does not precisely match the treatment plan and dosage details for hypoglycemia. The SheepDoctor significantly improves both the accuracy and detail richness of its responses. It correctly identifies the use of codonopsis, aconite, and cinnamon for treating hypoglycemia and provides dosage information that fully aligns with the reference answer. This demonstrates that the knowledge graph effectively enhances the model's understanding and application of domain-specific knowledge, enabling it to generate more precise and reliable answers. In conclusion, the combination of domain fine-tuning and knowledge graph integration significantly enhances the model's performance in the sheep disease question-answering tasks. The SheepDoctor not only accurately identifies disease types but also provides detailed dosage information, highlighting the advantages of the knowledge graph in expanding the model's domain knowledge coverage and improving its entity understanding.

Table 8 shows the results of the qualitative analysis of the different models on an unseen problem. LLaMA2–13B-Chinese fails to directly answer the question on drug dosage in lambs and only provided general recommendations, indicating its difficulty in generating specific and accurate answers when domain knowledge is lacking. The fine-tuned model can classify lambs by age but did not provide accurate dosage information, suggesting that the fine-tuning significantly improved performance, but there is still room for improvement. In contrast, SheepDoctor generates complete and accurate responses covering drug dosages for all age segments, which are almost identical to the reference answers. This suggests that the knowledge graph significantly enhances the model's ability to understand and apply domain knowledge, enabling it to generate more accurate and reliable responses.

Table 7
The qualitative analysis of different models for a specific problem.

Model Type	Question	Reference Answer	Model Answer
LLaMA2–13B-Chinese	党参,附子,桂枝是用来治疗什么疾病的∞剂量是多少∞	党参10克,附子10毫克,桂枝1克可以煎汁100毫升, 每次5毫升进行灌服, 每天1次, 连续3天.用于治疗低血糖症.	党参,附子,桂枝是中药材, 用于治疗肝气不调,胃气不调,肝气肾不调等疾病.剂量根据病情而定, 一般建议在医生指导下使用.
Fine-tuned model			党参,附子,桂枝是中药, 用于治疗咳嗽,鼻炎等症状的.剂量为每千克体重50毫克, 每天口服一次.
SheepDoctor			党参,附子,桂枝是用于处理低血糖症的.建议的剂量是党参10克,附子10毫克和桂枝1克100毫升, 每次用5毫升进行灌服.

Table 8
The qualitative analysis of different models for an unseen problem.

Model Type	Question	Reference Answer	Model Answer
LLaMA2–13B-Chinese	不同日龄羔羊单纯性消化不良的药物用量分别是多少∞	10日龄以内的羔羊, 每次用量为5~6ml;11~20日龄羔羊, 每次用量为8~10ml;21~30日龄羔羊, 每次用量为12~15ml.每日给药2~3次.	无法回答, 因为没有提供足够的信息来确定每个药物的正确剂量和用途.
Fine-tuned model			不同年龄的羔羊对药品的用量与其体重和发育阶段有关.一般, 新生羔羊的用量会比成年羊低, 而6月龄以上的羔羊用量可能比新生羔羊略高.具体用量应根据兽医的建议和药品说明书来确定.
SheepDoctor			10日龄内的羔羊的药物用量为5~6 ml, 21~30日龄内的羔羊的药量为12~15 ml, 11~20日龄内的羔羊的药量为8~10ml.

3.4. Comparison with other models

Since there is a relative lack of existing large models of agriculture, comparisons were made with generic models, in addition to comparing the effectiveness of SheepDoctor, answering effectiveness was also compared with generic models such as Chatgpt-4o and Kimi using 2034 Q&A data that had been constructed but not used for model fine-tuning. Table 9 shows the average performance of each model on these metrics. The differences in the performance of the models in the sheep disease Q&A task can be clearly seen.

Table 9 shows the performance comparison of different models in answering sheep disease questions. From the evaluation metrics,

Table 6
Performance of different models on unseen problems.

Model Type	BLEU-1 (%)	BLEU-2 (%)	BLEU-3 (%)	BLEU-4 (%)	ROUGE-1 (%)	ROUGE-2 (%)	ROUGE-L (%)	BERTScore (%)	Number-Units-Matched (%)
LLaMA2–13B-Chinese	30.47	24.31	21.31	18.82	51.21	4.89	50.68	68.44	67.53
Fine-tuned Model	32.69	25.67	22.11	19.28	56.94	4.31	56.56	69.49	71.43
SheepDoctor	48.12	42.67	38.87	35.44	73.99	28.88	72.88	78.14	76.19

Table 9
Different models for answering sheep disease questions.

Model Type	BLEU −1(%)	BLEU −2(%)	BLEU −3(%)	BLEU −4(%)	ROUGE −1(%)	ROUGE −2(%)	ROUGE -L(%)	BERT-Score(%)	Number-Units-Matched(%)
Kimi	8.44	5.34	3.71	2.74	0.74	0.07	0.74	63.65	85.39
ChatGPT-4o	9.49	5.91	3.89	2.74	0.90	0.25	0.86	63.90	85.50
SheepDoctor	23.17	15.92	12.00	9.32	14.81	7.98	14.17	68.64	89.82

SheepDoctor significantly outperforms the generic models Kimi and ChatGPT-4o in all metrics. Specifically, SheepDoctor scores much higher than Kimi and ChatGPT-4o in the BLEU series of metrics, which indicates that it has an obvious advantage in the fluency and accuracy of the generated text. Meanwhile, SheepDoctor's excellent performance on the ROUGE series of metrics further validates its ability in key information matching and semantic consistency. In addition, SheepDoctor's significant strengths in BERTScore and Number-Units-Matched metrics indicate that it is more capable of semantic understanding and entity matching.

To further analyze the effectiveness of the models in answering questions, we plotted Fig. 3, which shows the distribution of F1 values of the different models when answering the sheep disease question. This visualization allows for a more intuitive comparison of the stability and consistency of the models in terms of performance.

According to the distribution of F1 values in Fig. 3, Llama2 performs poorly in the sheep disease task, with 78.31 % of F1 values between 0.5 and 0.7 and only 0.05 % above 0.8, indicating its lack of domain knowledge. The GPT and Kimi models show similar distributions, with the majority of F1 values ranging from 0.6 to 0.7, accounting for 56.83 % and 41.62 % respectively, while only 2.02 % and 1.73 % of values exceed 0.8. In contrast, the fine-tuned model performs better, with 73.49 % of F1 values between 0.6 and 0.8 and 6.74 % above 0.8. The performance of SheepDoctor is further improved, with 73.63 % of F1 values between 0.6 and 0.8 and 8.26 % above 0.8, demonstrating the effectiveness of knowledge graph in enhancing the model's performance.

In summary, SheepDoctor performes well in the sheep disease quizzing task, especially on the high-precision task, a result that suggests that domain-specific model optimization and knowledge augmentation are effective ways to improve the performance of agricultural quizzing systems.

4. Discussion

This study introduces SheepDoctor, a large language model specifically designed for sheep disease diagnosis. By integrating domain-specific fine-tuning with a knowledge graph, SheepDoctor demonstrates superior performance in text generation fluency, key information matching, and semantic understanding. The model provides a reliable tool for intelligent diagnosis in agriculture, significantly improving diagnostic efficiency and accuracy. This research highlights the potential of combining domain-specific knowledge with large language models to address real-world challenges in specialized fields.

Although this study proves the effectiveness of SheepDoctor in the field of sheep disease diagnosis, the model still has some risk of error in practical application. Due to the high degree of similarity in the clinical symptoms of different sheep diseases, when the user expresses the symptoms vaguely, it is difficult for the model to accurately locate the corresponding diseases during the knowledge graph triad matching process, which can easily lead to confusion between different diseases. In addition, the completeness of user input information has an important impact on the accuracy of model inference. When the input information is insufficient or the symptom description is one-sided, the model lacks sufficient basis and is prone to miss diagnosis in the reasoning process. Especially when dealing with complex cases with multiple comorbidities, incomplete inputs increase the difficulty of model inference and further increase the risk of misdiagnosis.

In addition to the above analysis of the possible errors of the model,

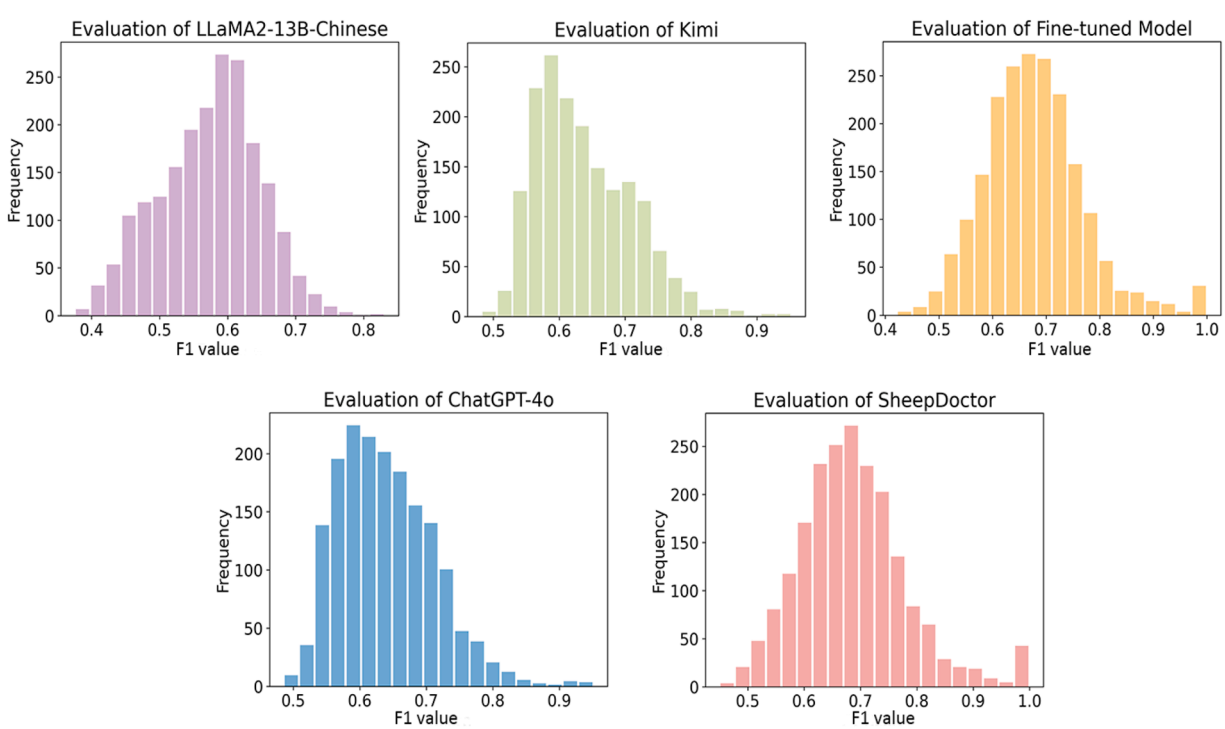


Fig. 3. Distribution of F1 values of different model.

SheepDoctor also has certain limitations. For example, the diversity and timeliness of the training data are insufficient. The existing training data mainly comes from limited Chinese literature resources, covering relatively limited regions and disease types, resulting in limited generalization ability of the model when dealing with other language scenarios or encountering novel diseases. Due to the risk of hallucination in the large model itself, SheepDoctor may experience hallucinations in the diagnosis of complex diseases, generating inaccurate information leading to misdiagnosis. In addition, in clinical practice, SheepDoctor is mainly used to assist decision-making, and its direct use without professional review may involve relevant ethical issues and harm to animal health.

Future research will address the limitations of SheepDoctor to improve it. The diversity, size, and timeliness of the training dataset will be further extended in the future, especially by introducing data from multiple languages, regions, and rare diseases to improve the generalization ability of the model. Meanwhile, the knowledge graph will be regularly updated to ensure the accuracy and timeliness of diagnostic information. To address the current problem that the model is prone to misdiagnosis in the case of ambiguous symptom descriptions and incomplete user input information, the introduction of a user input prompt template mechanism can be explored in the future to enhance the robustness of the model in an incomplete information environment. In clinical applications, although SheepDoctor can effectively assist in the preliminary diagnosis of sheep diseases, professional veterinary personnel are still required to review the results when dealing with complex or high-risk cases, in order to reduce the risk of hallucinations and safeguard animal health and safety. In addition, exploring the fusion of multimodal information such as image, video and sensor data is expected to further improve the diagnostic accuracy of the model and expand its potential for application in other professional fields. These improvements will help promote the development of intelligent agricultural diagnostics and expand the application prospects of SheepDoctor.

5. Conclusion

This study introduces SheepDoctor, a large language model integrated with a knowledge graph for sheep disease diagnosis. By constructing a sheep health question-answering dataset and fine-tuning LLaMA2–13B-Chinese with knowledge graph integration, the model's response generation capability is significantly enhanced. Experimental results demonstrate that SheepDoctor achieves high scores across BLEU, ROUGE and BERTScore, providing consistent and reliable diagnostic outcomes, along with accurate dosage and usage information for treatments, thereby increasing user trust. However, limitations remain, including limited data diversity and high computational resource requirements, which hinder its generalization ability and deployment in resource-constrained environments. Future work will focus on expanding dataset diversity and timeliness, optimizing model efficiency to reduce computational costs, and exploring the integration of additional domain knowledge to promote broader applications in agriculture.

Ethical statement

The research was conducted in accordance with ethical standards to ensure animal welfare and privacy protection. All data used in this study were anonymized, and no personally identifiable information was included. The dataset consists of anonymized health records obtained from veterinary clinics, with explicit consent for their use in research. No live animals were harmed during the study, and all procedures followed standard veterinary practices.

CRediT authorship contribution statement

Jiayi Xiong: Writing – original draft, Methodology, Investigation, Data curation. **Yong Zhou:** Resources, Funding acquisition. **Fang Tian:** Validation, Supervision, Resources, Conceptualization. **Fuchuan Ni:** Investigation, Funding acquisition. **Liang Zhao:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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